SCIENCE 1.6 **Name:** ***ANSWERS***

WORKSHEET ONE **DISTANCE, TIME, SPEED AND ACCELERATION**

1. Complete the following chart.

Quantity	Quantity symbol	Unit name	Unit symbol
Distance	***d***	metre	***m***
Mass	m	***kilogram***	***kg***
Time	***t***	***second***	s
Speed	***v***	***metres per sceond***	ms^{-1}
Acceleration	a	***metres per second²***	ms^{-2}

2. Carry out the following unit conversions. The first one is done for you.

(a) 10m = ***10,000*** mm
(b) 10min = ***600***s
(c) 1hour = ***3600*** s
(d) 600g = ***0.6*** kg
(e) 2.5km = ***2500*** m
(f) 950m = ***0.950*** km
(g) 4.3kg = ***4,300*** g
(h) 1.5hours = ***5400*** s
(i) 2.5min = ***150***s
(j) 500mm = ***0.500***m

3. (a) An airplane flies 5000 m in 20 s. What is its speed in ms^{-1} ?

$\mathbf{v} = d/t = 5000/20 = \mathbf{250\ ms^{-1}}$

(b) A car goes from Nelson to Greymouth (305 km) in 4.5 hours. What is its average speed in kmh^{-1}?

$\mathbf{v} = d/t = 305/4.5 = \mathbf{67.8\ kmh^{-1}}$

(c) A cyclist travels 1500 m in 4 min. What is his speed in ms^{-1}?

$\mathbf{v} = d/t = 1500/240 = \mathbf{6.25\ ms^{-1}}$

(d) A journey from Napier to Wellington is made in three sections.

> First section: Napier to Woodeville. Distance = 147 km. Time = 1.8 h.
> Second section: Woodeville to Masterton. Distance = 82 km. Time = 0.9 h.
> Third section: Masterton to Wellington. Distance = 102 km. Time = 1.7 h.

Calculate the speed of travel over each stage of the journey and the average speed of the whole journey.

First section: $\mathbf{v} = 147/1.8 = \mathbf{81.7\ kmh^{-1}}$

Second section: $\mathbf{v} = 82/0.9 = \mathbf{91.1\ kmh^{-1}}$

Third section: $\mathbf{v} = 102/1.7 = \mathbf{60\ kmh^{-1}}$

Average Speed = $\frac{147 + 82 + 102}{1.8 + 0.9 + 1.7}$ = **75.2 kmh^{-1}**

4. (a) A seagull flies at 10 ms^{-1} for 5 min. How far does it fly?

d = v x t = 10 x 300 = 3000m

(b) A snail crawls a distance of 3.8 cm at a speed of 0.002 ms^{-1}. How long does it take to do this?

$t = d/v = 3.8/0.002$ = ***1900s (= 31.7 min)***

(c) The average speed of a journey from Whangarei to Auckland is 84 kmh^{-1}. If the distance travelled is 170 km, how many hours did it take?

$t = d/v = 170/84$ = ***2.0 hours***

5. Use the speed formula to solve and complete the following chart.

Speed (ms^{-1})	12	***8.3***	50	0.8	5	***3.3***
Distance	***132 m***	500 m	1.4 km	***20 m***	100 m	2 km
Time	11 s	1 min	***28s***	25 s	***20 s***	10 min

6. A book falls off a shelf and accelerates towards the ground.
What is its acceleration? ***10 ms^{-2}***
How fast will it be going after one second? ***10 ms^{-1}***

7. A car accelerates from rest and reaches a speed of 15 ms^{-1} after 3 s. What is its acceleration?

$a = \Delta v/\Delta t = 15/3$ = **5 ms^{-2}**

8. A rocket is fired into the air and its upward speed is 13 ms^{-1} as it clears the top of a nearby building. Three seconds later its upward speed is 25 ms^{-1}. What is its acceleration?

$a = \Delta v/\Delta t = 12/3$ = **4 ms^{-2}**

9. A girl, starting from rest, skis down a mountain slope and after six seconds is travelling at a speed of 14 ms^{-1} What is her acceleration?

$a = \Delta v/\Delta t = 14/6$ = **2.3 ms^{-2}**

10 Using the equation for acceleration complete the following chart.

Acceleration	2.4 ms^{-2}	***7.3*** **ms^{-2}**	0.8 ms^{-2}	18 ms^{-2}	***4*** **ms^{-2}**	30 ms^{-2}
Change in speed	60 ms^{-1}	44 ms^{-1}	***96*** **ms^{-1}**	120 ms^{-1}	60ms^{-1}	***105*** **ms^{-1}**
Time	***25 s***	6 s	2 min	***6.7 s***	15 s	3.5 s

11. A car is travelling at 5 ms^{-1} and accelerates at 4 ms^{-2} for 6 s. What is its final speed?

$\triangle$v = a x $\triangle$t = 4 x 6 = 24 ms^{-1}
Final speed = 5 + 24 = 29 ms^{-1}

12. A cyclist travelling at 3 ms^{-1} accelerates at 0.5 ms^{-2}. What will her speed be after 4 s?

$\triangle$v = a x $\triangle$t = 0.5 x 4 = 2 ms^{-1}
Final speed = 3 + 2 = 5 ms^{-1}